ABS INDICATOR

DESCRIPTION

Fig. 7 ABS Indicator

An Antilock Brake System (ABS) indicator is standard equipment on all instrument clusters (Fig. 7). However, this indicator is only functional on vehicles equipped with the ABS option. The ABS indicator is located near the lower edge of the instrument cluster, to the left of center. The ABS indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Failure of Anti-lock Braking System” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The ABS indicator is serviced as a unit with the instrument cluster.

OPERATION

The ABS indicator gives an indication to the vehicle operator when the ABS system is faulty or inoperative. This indicator is hard wired on the instrument cluster electronic circuit board, and is completely controlled by the Controller Antilock Brake (CAB). The ABS indicator Light Emitting Diode (LED) receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions; therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the CAB through the CAB relay in the Power Distribution Center (PDC). The CAB will turn on the ABS indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the ABS indicator is illuminated by the cluster for about two seconds as a bulb test.
- **ABS Self-Test Fault** - Each time the CAB detects a fault in a monitored ABS circuit, the ABS indicator will be illuminated. The indicator remains illuminated until the fault condition is corrected, or until the ignition switch is turned to the Off position, whichever occurs first.
- **ABS Diagnostic Test** - The ABS indicator is blinked on and off by the CAB during the performance of the ABS diagnostic tests.

The CAB continually monitors the ABS circuits and sensors to decide whether the system is in good operating condition. If the CAB turns the ABS indicator on after the bulb test, it indicates that the CAB has detected a system malfunction and/or that the ABS system has become inoperative. The CAB will store a Diagnostic Trouble Code (DTC) for any malfunction it detects. (Refer to 5 - BRAKES - DESCRIPTION). The ABS indicator can be diagnosed using conventional diagnostic tools and methods.

DIAGNOSIS AND TESTING - ABS INDICATOR

The diagnosis found here addresses an inoperative Antilock Brake System (ABS) indicator condition. If there are problems with several indicators in the instrument cluster, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the ABS indicator stays on with the ignition switch in the On position or comes on and stays on while driving, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If no ABS problem is found, the following procedure will help to locate a short or open in the ABS warning indicator driver circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C2) for the instrument cluster from the connector receptacle on the back of the cluster housing. Reconnect the battery negative cable. Turn the ignition switch to the On